What is Big Data?

Big data refers to large datasets that are challenging to store, search, share, visualize, and analyze.

"***Big Data***" is data whose scale, diversity, and complexity require new architecture, techniques, algorithms, and analytics to manage it and extract value and hidden knowledge from it...

What is Big Data? (3V)

- *High volume, velocity, and /or variety information assets that demand new, innovative forms of processing for enhanced decision making, business insights or process optimization*

Why Big Data?

Advances in Computing Technologies

↓

Sudden Explosion of data

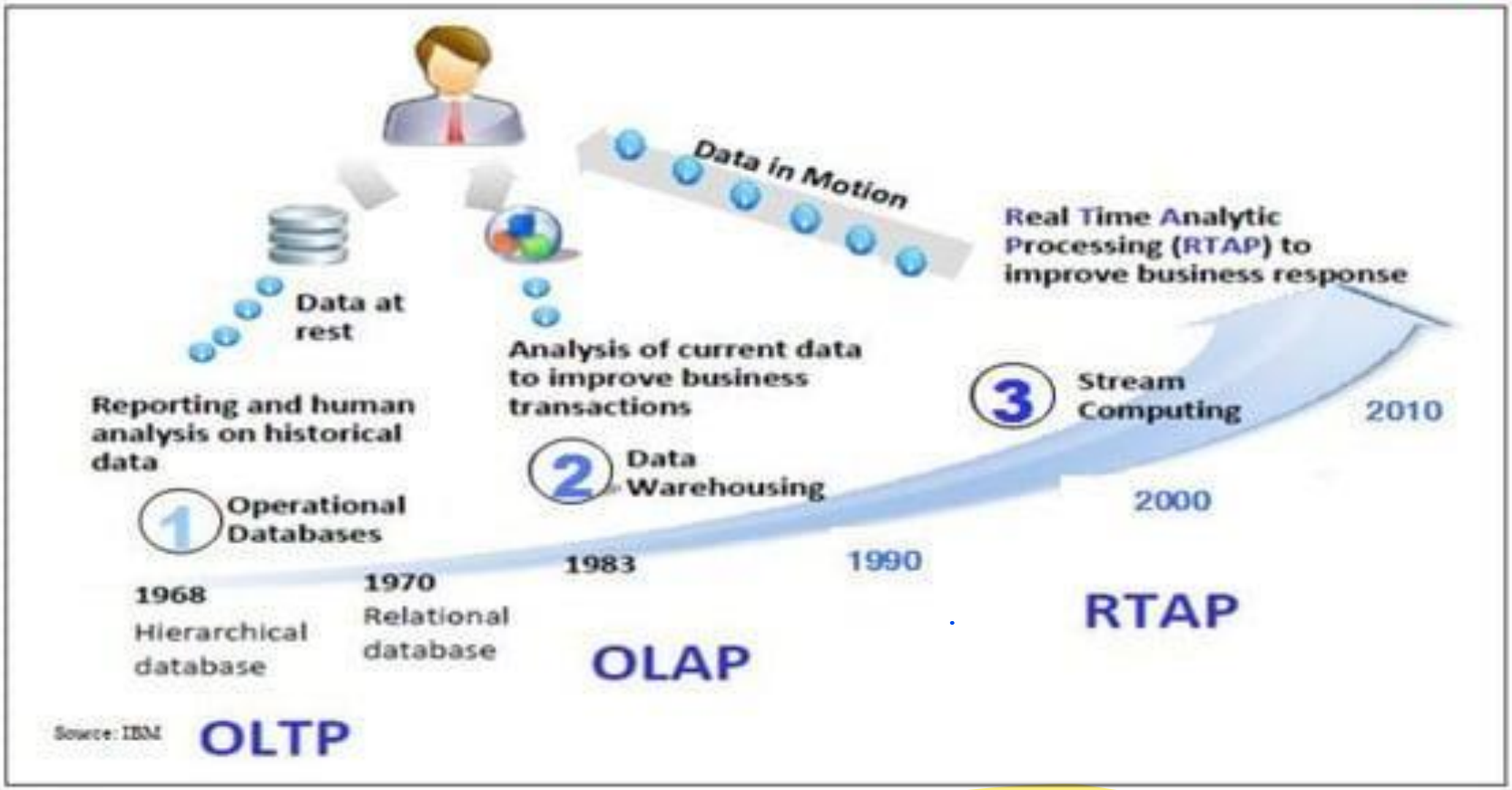
↓

Emergence of Data-Driven Paradigm

↓

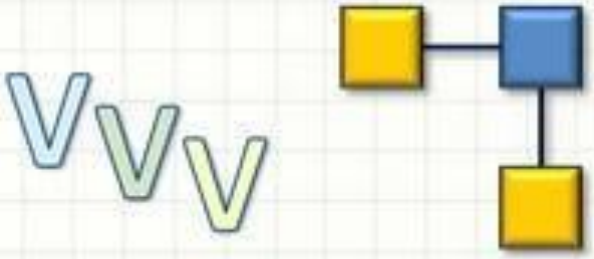
Big Data Technologies

Harnessing Big Data

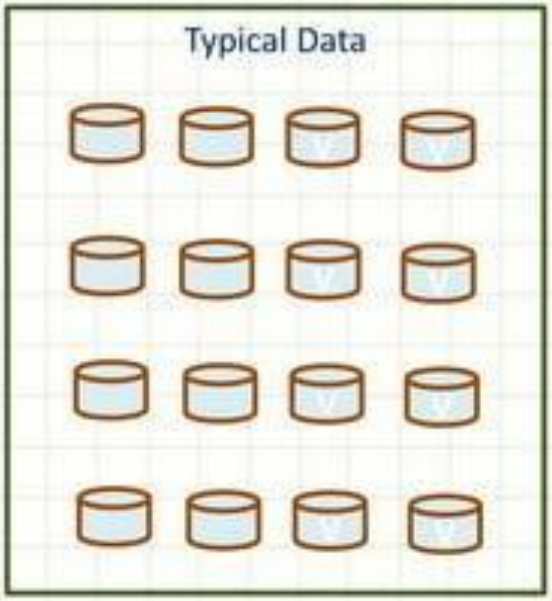


- **OLTP:** Online Transaction Processing (DBMSs)
- **OLAP:** Online Analytical Processing (Data Warehousing)
- **RTAP:** Real-Time Analytics Processing (Big Data Architecture & technology)

Big Data



Typical Data

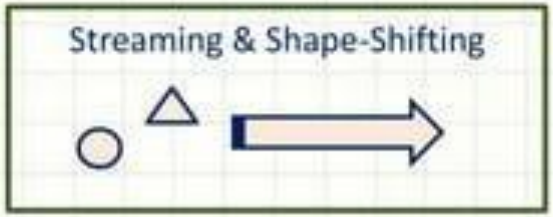


Big Data

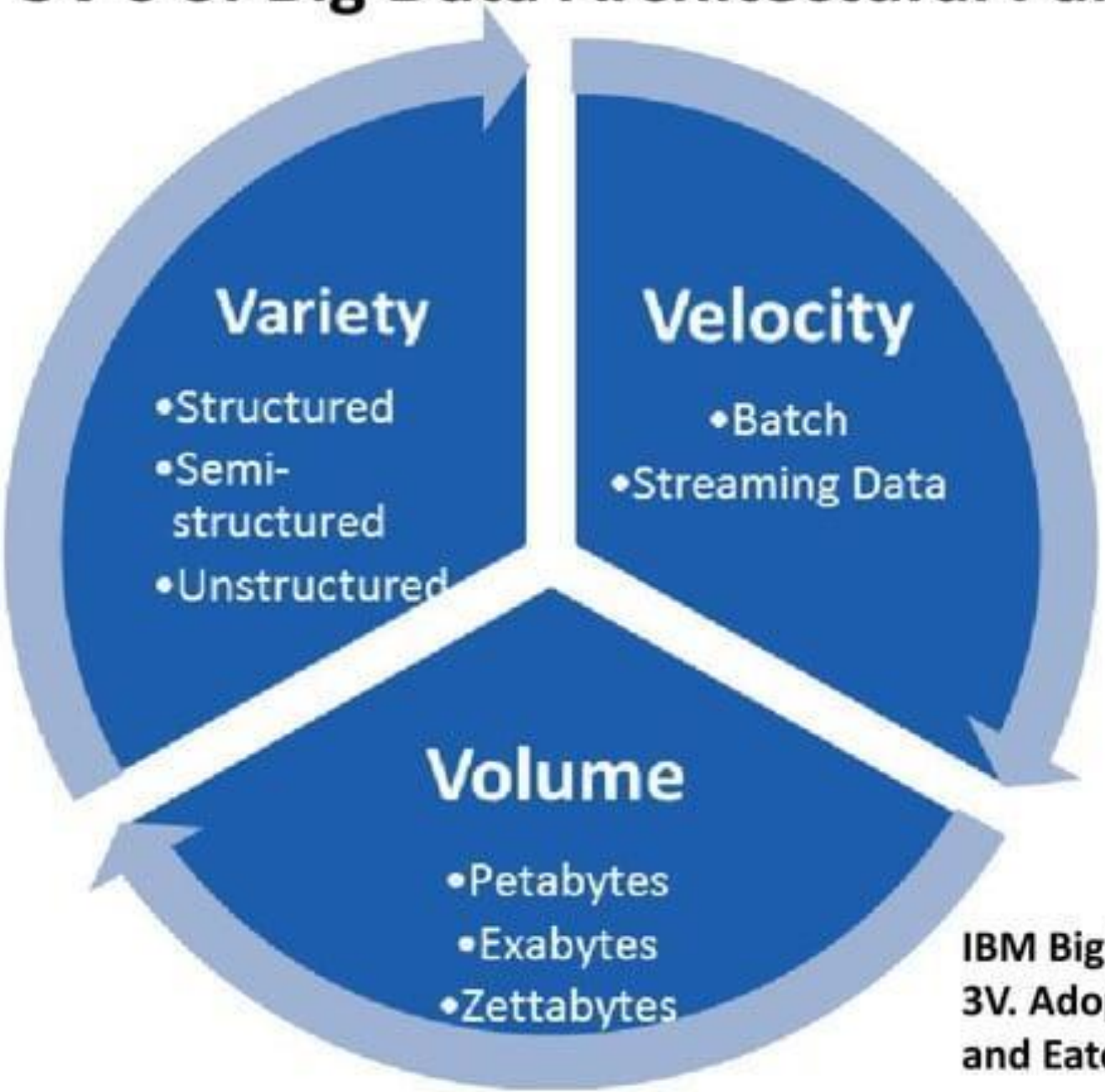
A

B

C



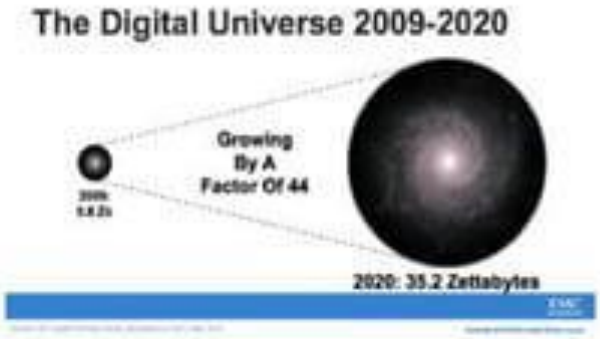
3V's of Big Data Architectural Paradigms

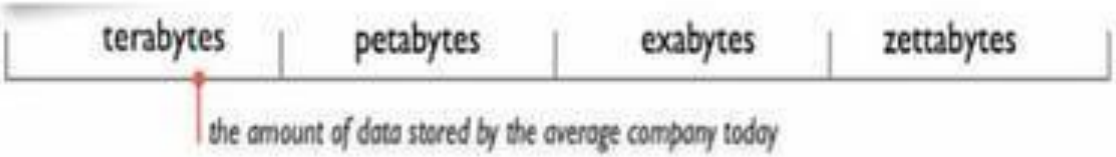


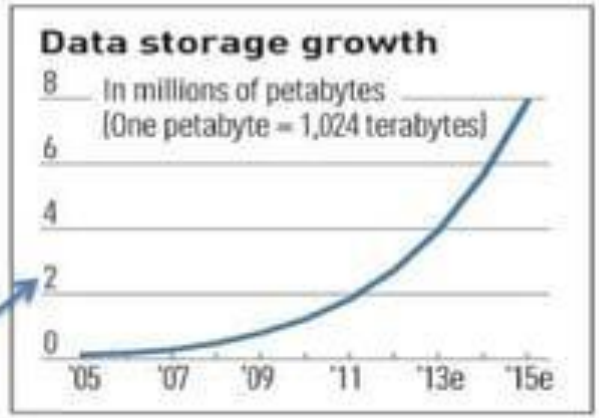
IBM Big Data characteristics – 3V. Adopted from (Zikopoulos and Eaton 2011)

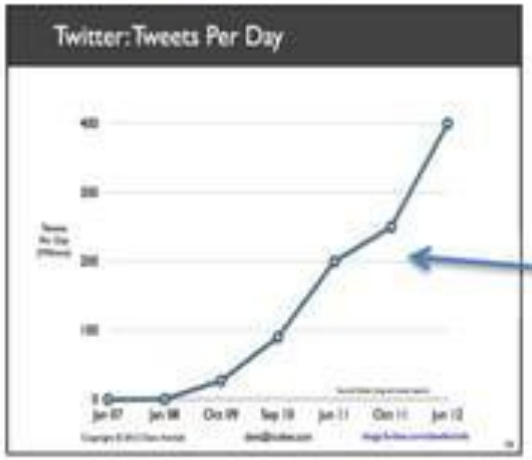
Volume (Scale)

- **Data Volume**
  - 44x increase from 2009 2020
  - From 0.8 zettabytes to 35zb
- Data volume is increasing exponentially







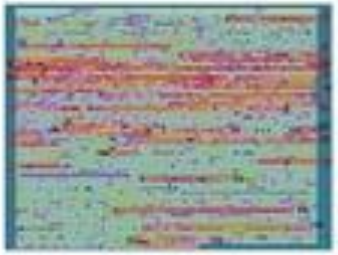


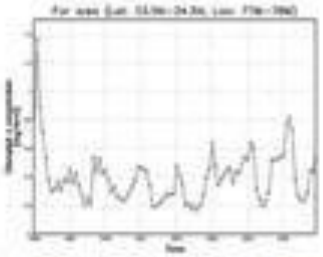
Exponential increase in collected/generated data

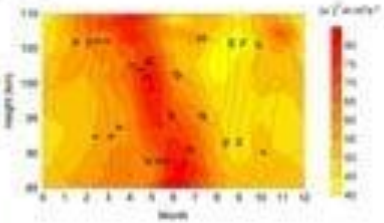
12+ TBs
of tweet data
every day
? TBs of
data every day
25+ TBs of
log data
every day
30 billion RFID
tags today
(1.3B in 2005)
4.6
billion
camera
phones
world wide
100s of
millions
of GPS
enabled
devices sold
annually
2+
billion
people on
the Web
by end
2011
76 million smart meters
in 2009...
200M by 2014
Google
talk
Google
Google Analytics
YouTube
http://www.

Variety (Complexity)

- Relational Data (Tables/Transaction/ Legacy Data)
- Text Data (Web)
- Semi-structured Data (XML)
- Graph Data
  - Social Network, Semantic Web (RDF), …
- Streaming Data
  - You can only scan the data once
- A single application can be generating/collecting many types of data
- Big Public Data (online, weather, finance, etc)







To extract knowledge → all these types of data need to linked together

Types of Big Data

- **Structured**
  - Data that can be stored and processed in a fixed format, aka schema
- **Semi-structured**
  - Data that does not have a formal structure of a data model, i.e. a table definition in a relational DBMS, but nevertheless it has some organizational properties like tags and other markers to separate semantic elements that makes it easier to analyze, aka XML or JSON
- **Unstructured**
  - Data that has an unknown form and cannot be stored in RDBMS and cannot be analyzed unless it is transformed into a structured format is called as unstructured data
  - Text Files and multimedia contents like images, audios, videos are example of unstructured data - unstructured data is growing quicker than others, experts say that 80 percent of the data in an organization is unstructured

Velocity (Speed)

- Data is begin generated fast and need to be processed fast
- Online Data Analytics
- Late decisions ➔ missing opportunities

- **Examples**
  - **E-Promotions:** Based on your current location, your purchase history, what you like ➔ send promotions right now for store next to you
  - **Healthcare monitoring:** sensors monitoring your activities and body ➔ any abnormal measurements require immediate reaction

Real-time/Fast Data

Social media and networks
(all of us are generating data)

Scientific instruments
(collecting all sorts of data)

Mobile devices
(tracking all objects all the time)

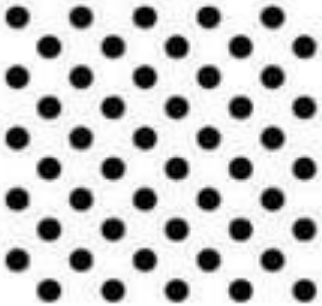
Sensor technology and networks
(measuring all kinds of data)

- The progress and innovation is no longer hindered by the ability to collect data
- But, by the ability to manage, analyze, summarize, visualize, and discover knowledge from the collected data in a timely manner and in a scalable fashion

Real-Time Analytics/Decision Requirement

Some Make it 4V's

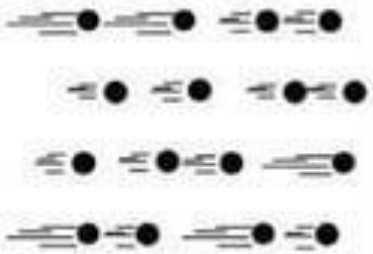
Volume



Data at Rest

Terabytes to exabytes of existing data to process

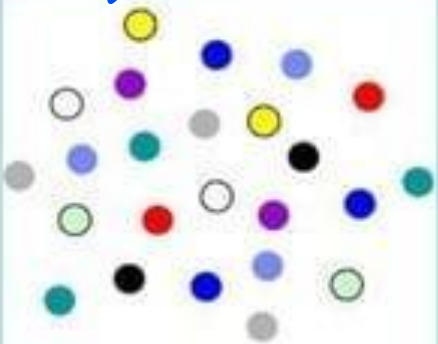
Velocity



Data in Motion

Streaming data, milliseconds to seconds to respond

Variety



Data in Many Forms

Structured, unstructured, text, multimedia

Veracity*

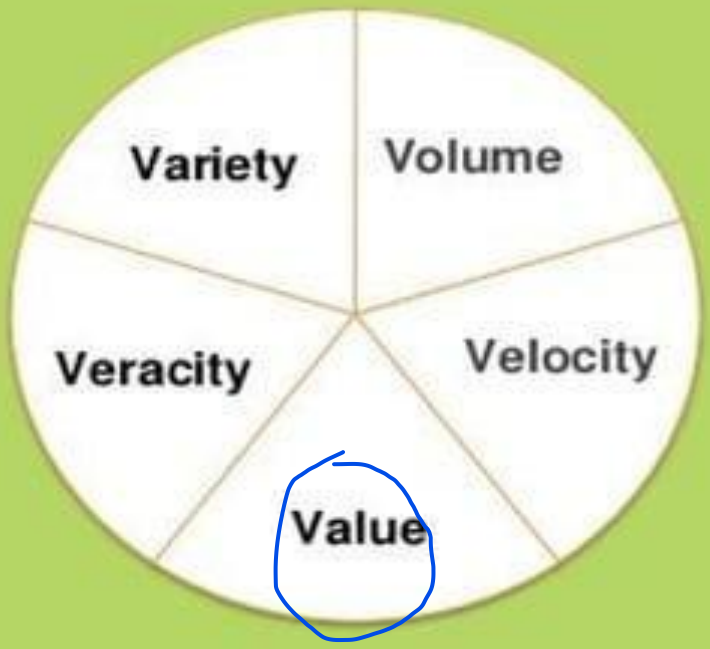
Data in Doubt

Uncertainty due to data inconsistency & incompleteness, ambiguities, latency, deception, model approximations

Some Make it 5 V's

To get a better understanding of what Big Data is, it is often described using

5 Vs:



BROUGHT TO YOU BY THE BESTSELLING AUTHOR OF...

Big Data

- Volume

  Huge Volumes of Data

- Velocity

  Drinking from a Fire Hose

- Variety

  n-Structured Data

- Veracity

  Quality, Accuracy, Reliability, Trustworthiness

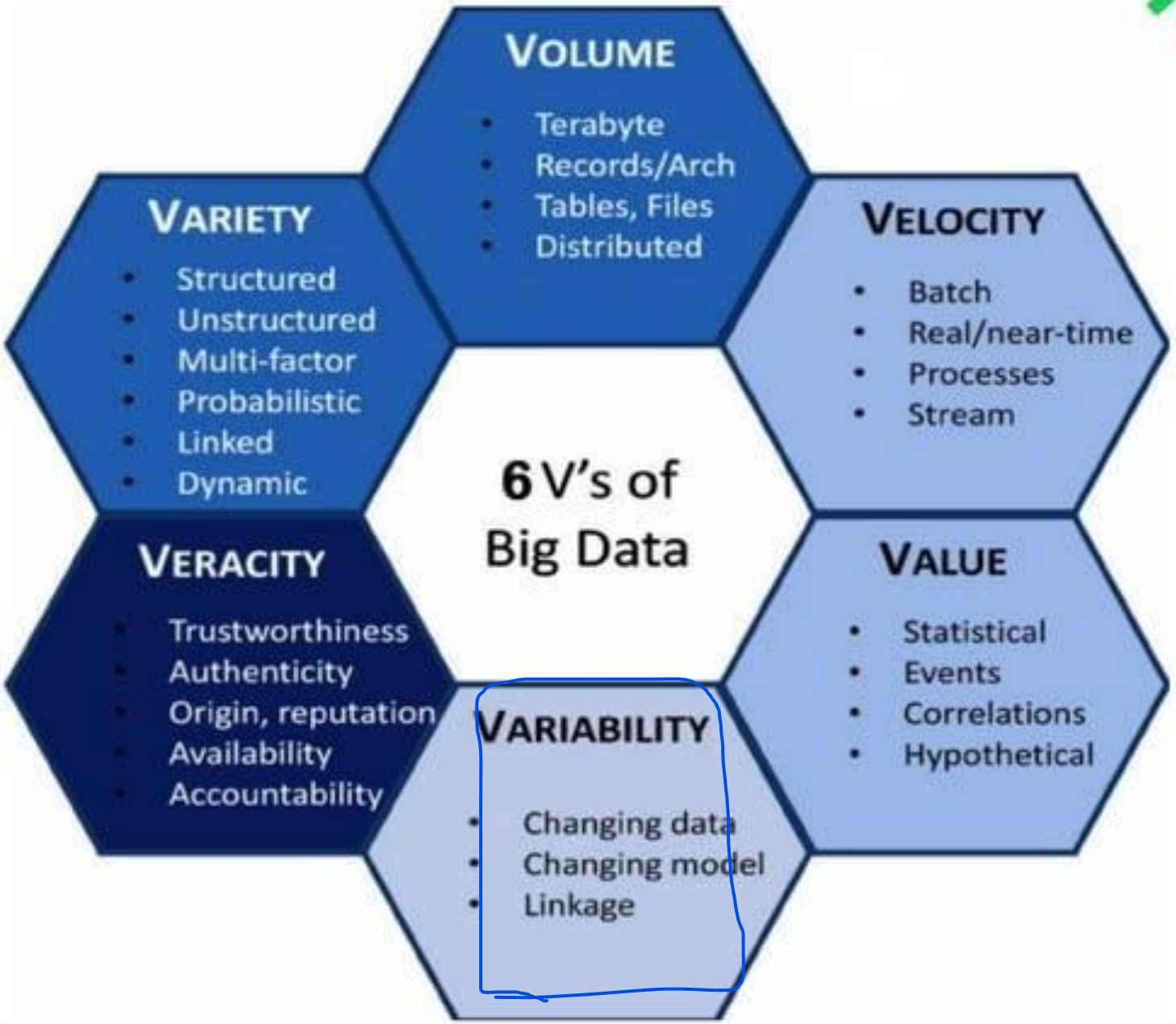
- Value

  Business Value and Value Potential

Value

- Value is defined as the usefulness of data for an enterprise.
- The value characteristic is intuitively related to the veracity characteristic in that the higher the data fidelity, the more value it holds for the business.
- Value is also dependent on how long data processing takes because analytics results have a shelf-life; for example, a 20 minute delayed stock quote has little to no value for making a trade compared to a quote that is 20 milliseconds old.
- Data that has high veracity and can be analyzed quickly has more value to business.

The 6 V's Big Data traits

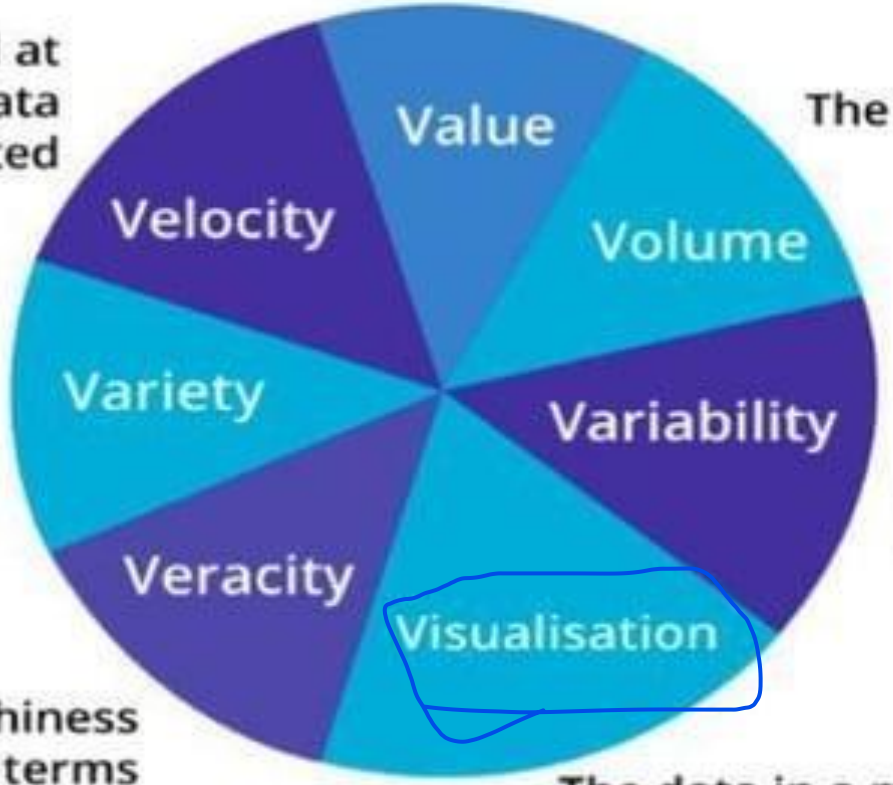


The 7 Vs OF BIG DATA

Just having Big Data is of no use unless we can turn it into value

The speed at which the data is generated

The size of the data



The different types of data

The data whose meaning is constantly changing

The trustworthiness of the data in terms of accuracy

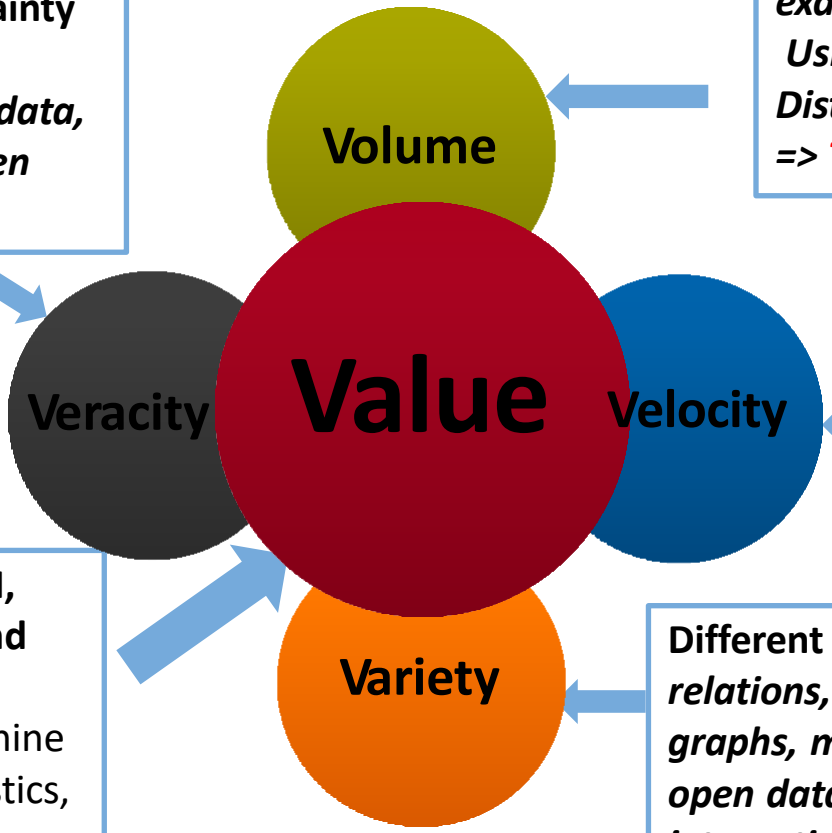
The data in a manner that's readable and accessible

The 9 V's Big Data traits

What is Big Data?

Large volume, cloud, HDFS, EDW, NoSQL
Scale: ***terabytes , petabytes, exabytes***
Using Commodity hardware, Distributed storage (HDFS)
=> Technology solution

Quality, reliability, uncertainty and meaning *in*
data itself (e.g., weather data, translation of hand-written data)



Speed to create/capture/process/ /store
Processing mode: Real-time; streaming; in-memory computing
=> Semi-Technology solution

Actionable knowledge, ROI, Relevancy to customers, and business value
Analysis: SQL queries, machine learning, data mining, statistics, visualization, optimization, decision analysis

Different data types and sources *(e.g., relations, documents, web, XML files, graphs, multimedia, IOT, dark data, open data, external data) -=>data integration/ETL/ELT/Data Virtualization*
=> SW solution

Use Cases in Big Data Projects

- **Marketing and sales growth**
- **Operational and financial performance improvement**
- **Risk and compliance management**
- **New product and service innovation**